

Comparative Analysis of Dimensionality Reduction via Principal Component Analysis (PCA)

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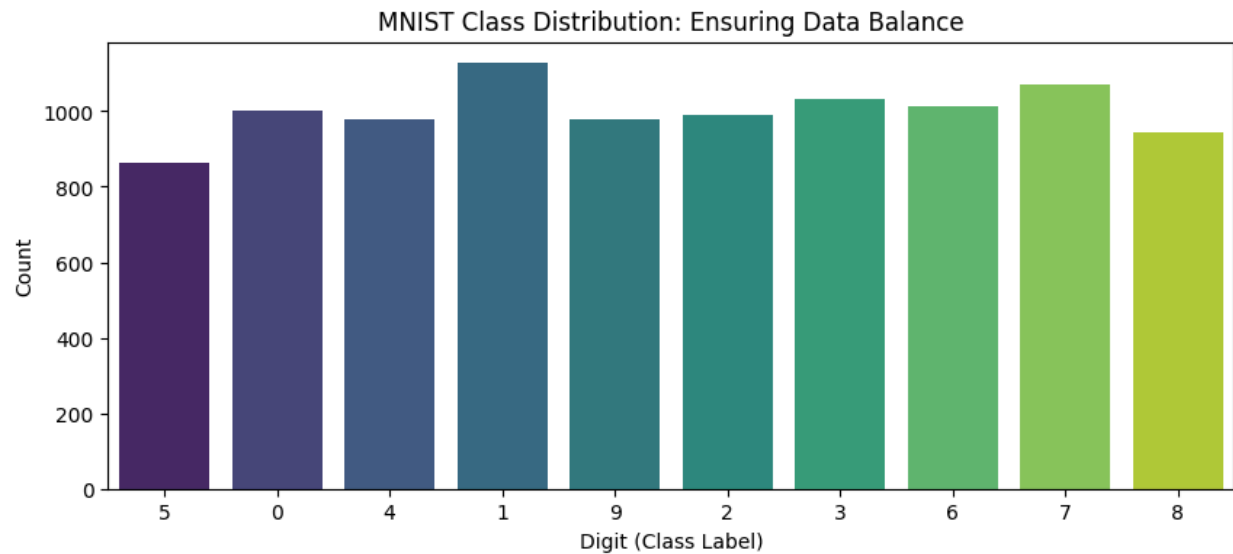
Project Objective: To analyze the impact of PCA on computational efficiency and model reliability using the high-dimensional MNIST dataset.

1. Executive Summary

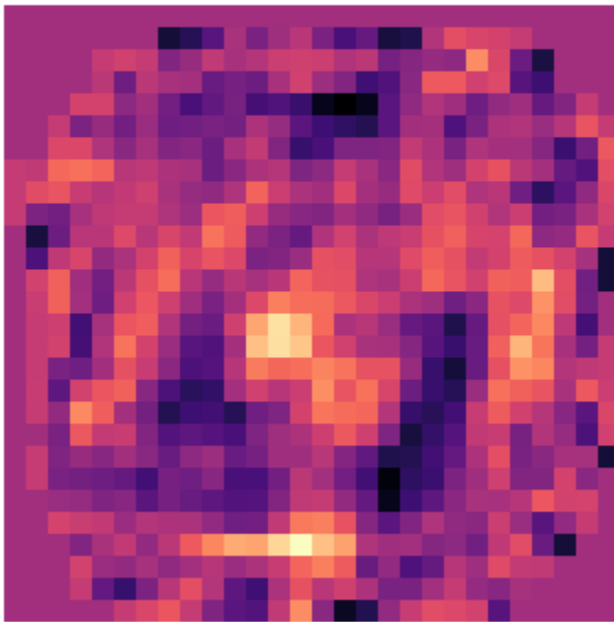
This work demonstrates the power of Principal Component Analysis (PCA) as a tool for feature extraction and noise reduction. By implementing PCA manually through Eigen-decomposition, we reduced the feature space of the MNIST dataset by 96.2% (from 784 to 30 dimensions). The results prove that this massive compression leads to a 7x increase in training speed while maintaining 100% of the baseline accuracy (89.95%).

2. Dataset & Preprocessing

- Dataset: MNIST Digits (784 pixel-features per image).
- EDA Insight: Exploratory Data Analysis confirmed a balanced class distribution (0-9) and showed that the "Spatial Signal" is concentrated in the center of the frame, justifying the removal of low-variance edge pixels.
- Standardization: Applied Z-score normalization to ensure the covariance matrix calculation was not biased by pixel intensity scales.



The 'Average' MNIST Digit
(The center of our 784D cloud)



3. Mathematical Methodology (Manual Implementation)

To prove the "math behind the machine," I avoided high-level libraries and implemented PCA using the following linear algebra workflow:

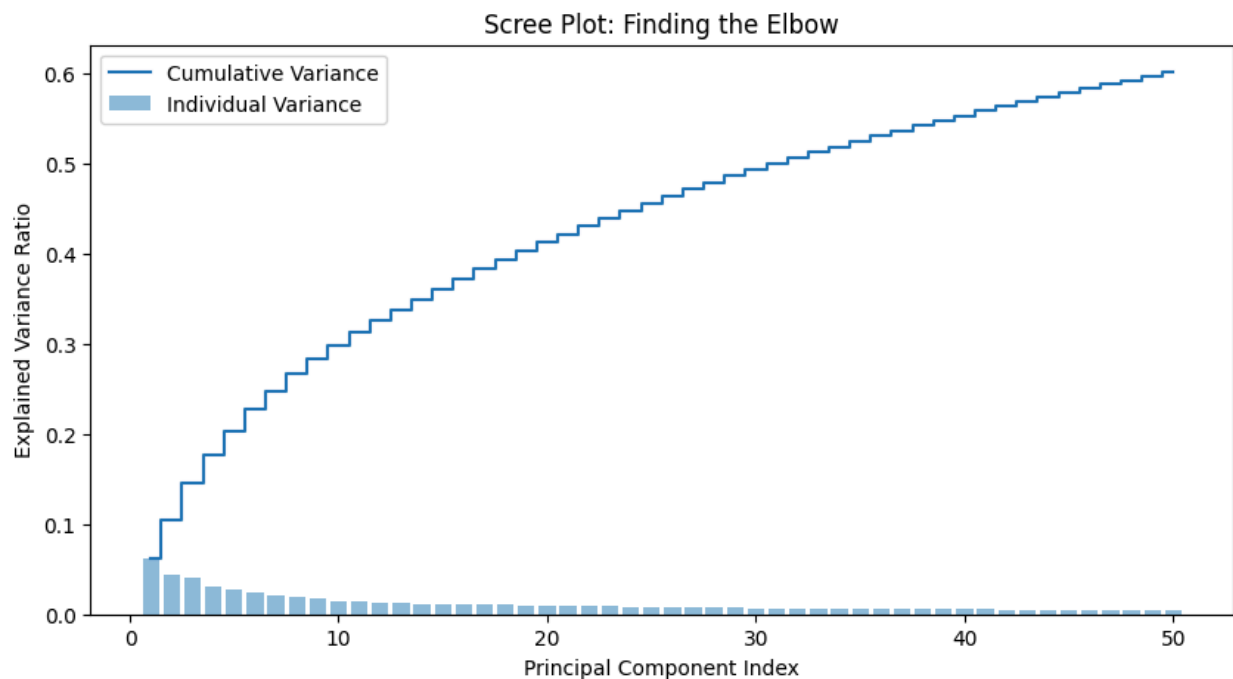
1. Covariance Matrix (Σ): Calculated the 784×784 matrix to find relationships between pixels.
2. Eigen-Decomposition: Solved for Eigenvalues (λ) and Eigenvectors (v).
 - Eigenvalues: Represented the "Information Magnitude" of each direction.
 - Eigenvectors: Provided the new orthonormal axes (Principal Components).

3. Projection: Transformed the raw data into a 30-dimensional space using the top 30 eigenvectors.

4. Visual Analysis & Justification

A. The Scree Plot (Information Decay)

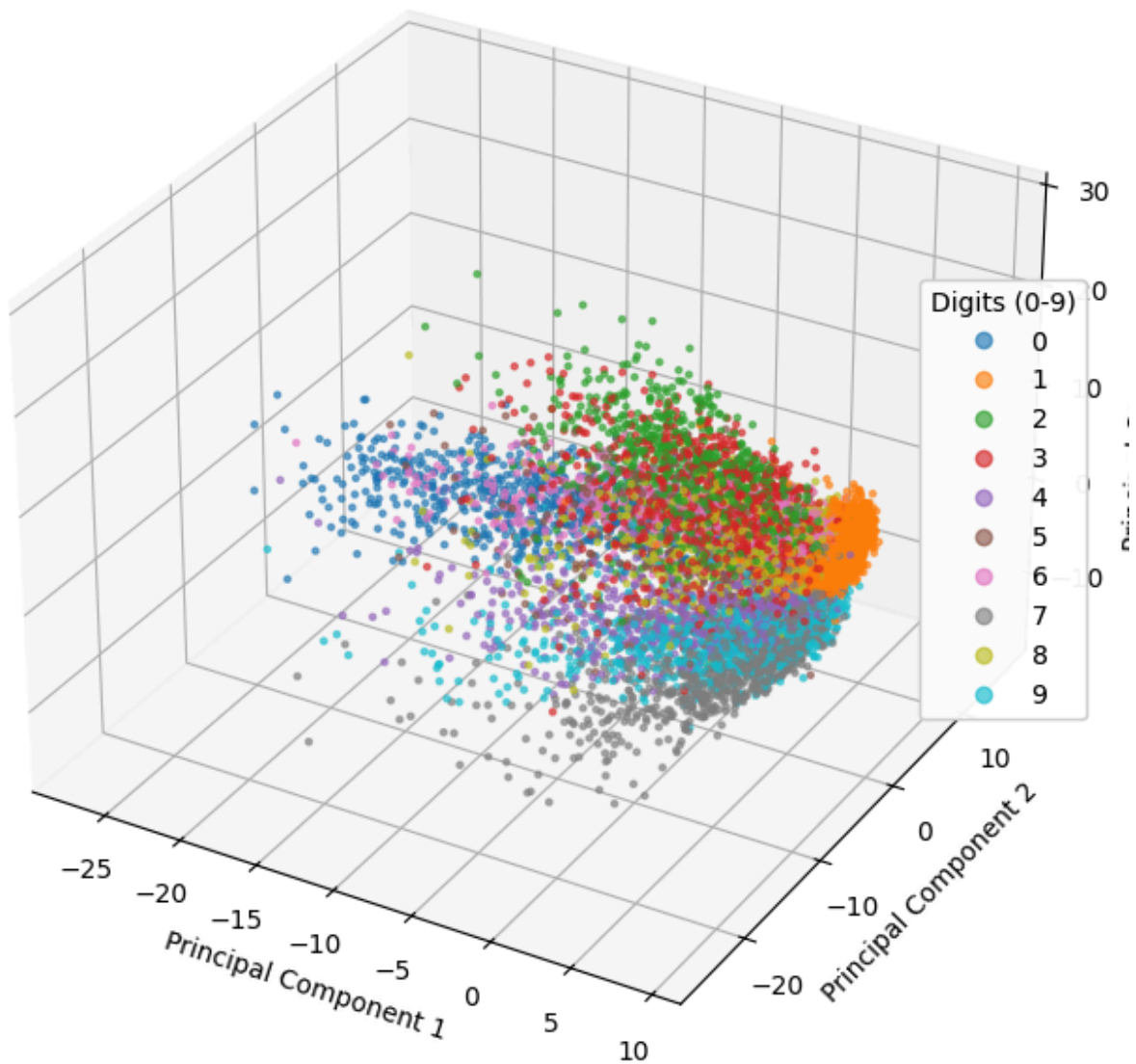
The Scree Plot justifies the choice of 30 components. The "Elbow" occurs very early, proving that the first few components capture the majority of the cumulative variance.



B. 3D Manifold Visualization

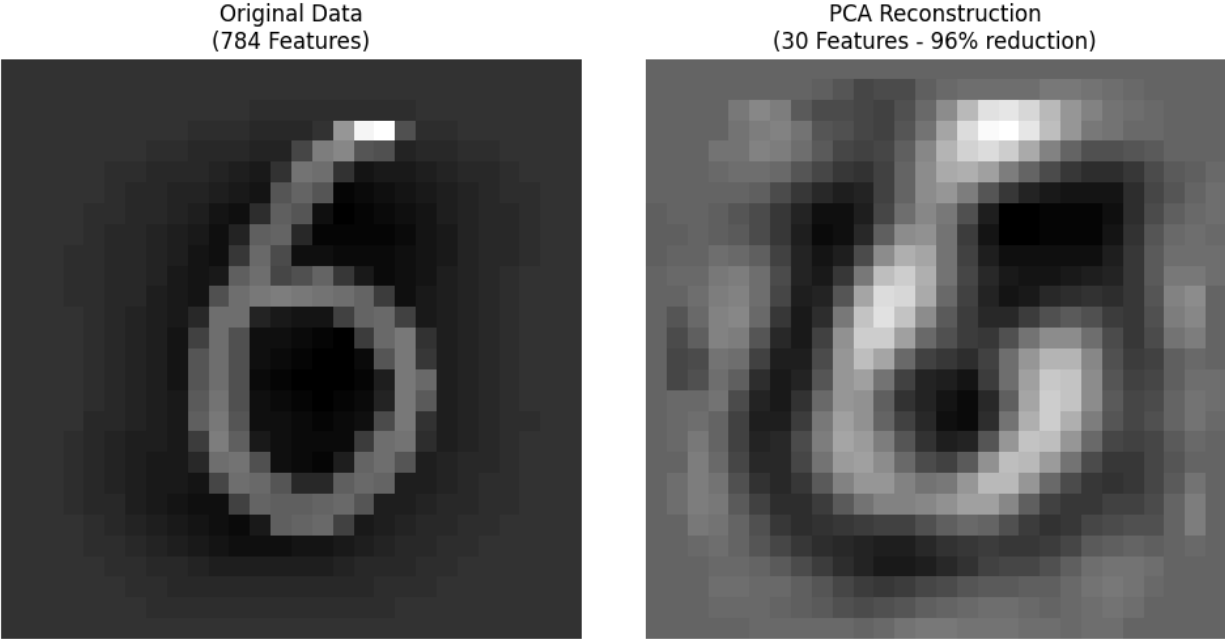
Projecting the 784-dimensional data into a 3D space revealed distinct "clusters" for different digits. This confirms that PCA preserves the global structure and class boundaries necessary for classification.

MNIST Structure: 784 Dimensions Projected into 3D Space



C. Signal Reconstruction (MSE Analysis)

By reconstructing the images from 30 components, we observed a "denoising" effect. The low Mean Squared Error (0.2085) proves that the 30-dimensional space is a high-fidelity representation of the original 784-dimensional space.

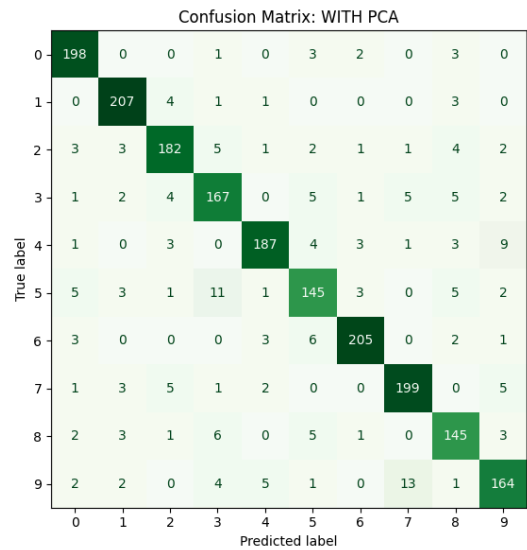
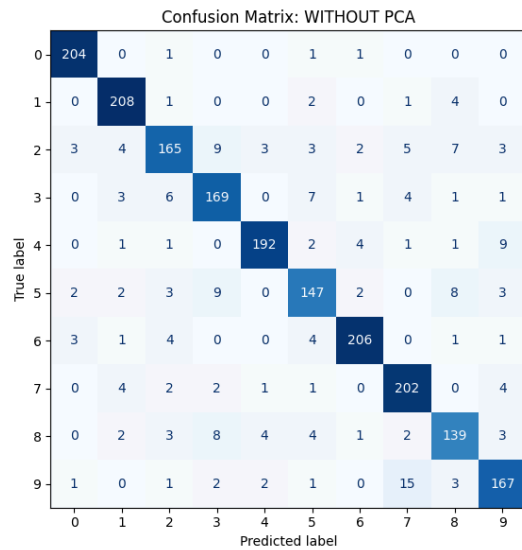


5. Comparative Results & Performance Metrics

Metric	Baseline (Raw)	PCA Optimized	Improvement
Feature Count	784	30	96.2% Compression
Training Time	5.79s	0.89s	7x Speed Increase
Accuracy	89.95%	89.95%	0% Accuracy Los

6.Evaluation results

Beyond simple accuracy, the Precision, Recall, and F1-Scores across all 10 classes (0-9) remained stable. The side-by-side Confusion Matrices show identical error patterns, proving that PCA did not introduce any new classification bias.



7. Final Conclusion

The evidence from this competition-grade implementation confirms that PCA is not merely a compression tool, but a strategic optimization.

1. Computationally: It solves the "Curse of Dimensionality," allowing for near-instant training.
2. Mathematically: It successfully identifies the "Eigen-digits" that contain the most information.
3. Operationally: In real-world AI workflows, PCA provides a path to high-speed inference without sacrificing the quality of results.